

RAHUL MITRA

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EDUCATION

National University of Singapore

Aug 2023 – Jul 2027

Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Operations Research (NUS System Design Project), Contract, STMicroelectronics

Aug 2026 – Present

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

Robotics Vision Engineer, Amazon (Blended AI+X)

Jul 2026 – Present

- Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 – Present

- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.

Platform & Solutions Engineer (Sparks Citizen Developer), People's Association

Sep 2025 – Sep 2026

- Delivered a digital garden-plot balloting platform across admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Led deployment planning, multi-repo coordination, and stakeholder management across product, engineering, and operations, winning the S\$20,000 Sparks Community Innovation Fund.

Data Analytics & Product Management Intern, Abbott Laboratories

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Productionized a team-built APC simulator for live internal users on Azure App Service with Docker packaging, access restrictions, and runtime and session hardening.

PROJECTS

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)

Jan 2026

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.

LEADERSHIP AND ACTIVITIES

YesWeHack, Independent Bug Bounty Researcher

May 2024 – Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.

NTUC Health Active Ageing Centre, Elderly Care Volunteer

Sep 2025 – Sep 2026

- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA

Web and Cloud: React, Next.js, Three.js, FastAPI, Flask, Docker, AWS (Fargate, Route 53), Terraform, Azure App Service, Redis, Kafka, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping

AI and Machine Learning: RAG, AI agents, vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), CUDA/HPC on ASPIRE 2A

Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)